DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-ITCS214
COURSE TITLE	**:**	Data Structures and Algorithms
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	2 Units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	S-ITCS123 / S-ITCS123LA
CO-REQUISITES	**:**	S-ITCS214LA
CONSULTATION HOURS	**:**	Saturday: 12:00 – 2:00 COS 100-C Friday : 1:00-3:00 ONLINE
CONTACT INFORMATION	**:**	sperolin@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course provides the concepts of abstract data types of common data structures such as lists, linked lists, stacks, queues, trees and graphs used in solving computer problems. This course also includes the study of time and space complexities of algorithms and the choice of appropriate data structures. The course covers several data structures as tools for programming, best practices in using those data structures with estimated running time and space efficiency, hands-on exercises to develop practical skills in creating impactful programs

COURSE LEARNING OUTCOMES:

By the end of this course, students are expected to:

CLO1. Understand the importance of data structures and algorithms in application of daily living.
CLO2. Evaluate the running time of algorithms and data structures using big-oh notations.
CLO3. Analyze the features of each data structure and its operations and use it to design algorithms to solve real-world problems.
CLO4. Apply the most appropriate data structures given a specified real - world problem.
CLO5. Create codes written in Java for various algorithms and data structures.
CLO6. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN FOR MIDTERM PERIOD:

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand data structures & algorithms in application of daily living.	**TLO1.** Distinguish the differences between algorithms and data structures. **TLO2.** Identify the criteria for efficient algorithms.
CLO2. Evaluate the running time of algorithms and data structures using big-oh notations.	**TLO3.** Calculate the running time of codes following specific rules.
CLO3. Analyze the features of each data structure and its operations and use it to design algorithms to solve real-world problems.	**TLO4.** Construct pseudo-language implementations of lists and its operations. **TLO5.** Write algorithms for linked list and its variations. **TLO6.** Understand the concept of a stacks and its operations.
CLO6. Demonstrate 21st century skills in all learning activities.	**TLO7.** Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem- Solving, Use of ICT (not part of topic, but integrated in assessment)

LEARNING OUTCOMES FOR MIDTERM PERIOD

Module Title and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy Module Activities Online/ Offline	Technology Enabler	Asynchronous	Onsite	Allotted Hours
Module 0 CLO1 TLO1	**W1** Aug 19 – 24	Gospel Reading, Reflection and Prayer Presentation of the course syllabus **Module 0 – Course Introduction** Introduction of the course professor Presentation of the Enrich Virtual Learning model LMS Formative Activity: Data Structures Primer https://www.youtube.com/watch?v=bum_19loj9A&list=PLBZBJbE_rGRV8D7XZ08LK6z-4zPoWzu5H	Classroom PowerPoint YouTube		✓	2
Module 1 CLO1 TLO2	**W2** Aug 26 – 31	Gospel Reading, Reflection and Prayer **Module 1 – Introduction to Data Structures** Class Participation Activity	Schoolbook PowerPoint Quizzes		✓	2
Module 2 TLO3, CLO2, CLO6	**W3** Sept 2 – Sept 7	Gospel Reading, Reflection and Prayer **Module 2 – Introduction to Algorithm Analysis** Class Participation Activity Enabling Assessment: Quiz om Algorithm Analysis	Schoolbook PowerPoint Canva	✓		2

Module 3 CLO2 TLO3	**W4** Sept 9 – Sept 14	Gospel Reading, Reflection and Prayer **Module 3 – List as an abstract data type** Class Participation/ Warm-Up Activity: Using Wooclap for Module 3 Enabling Assessment: Quiz on List as an Abstract Data Type	Schoolbook PowerPoint Wooclap		✓	2
Module 4 TLO5, CLO3, CLO5, CLO6	**W5** Sept 16 – Sept 21	Gospel Reading, Reflection and Prayer **Module 4 – Linked List as an abstract data type** Class Participation Activity for Module 4	Schoolbook Canva	✓		2
Module 4 TLO5, CLO3, CLO5, CLO6	**W6** Sept 23 – Sept 28	Gospel Reading, Reflection and Prayer **Module 4 – Linked List Applications** Class Participation/ Warm-Up Activity: Using Wooclap for Module 4	Schoolbook Canva	✓		2
Module 5 TLO6, CLO3, CLO5, CLO6	**W7** Sept 30 – Oct 5	Gospel Reading, Reflection and Prayer **Module 5: Stack as an abstract data type** Enabling Assessment: Quiz on Linked List Application and Stack as an Abstract Data Type	Schoolbook PowerPoint Wakelet Quizlet		✓	2
Module 5 TLO6, CLO3, CLO5, CLO6	**W8** Oct 7 – Oct 12	Gospel Reading, Reflection and Prayer **Module 5: Stack Applications – Prefix, Infix, Postfix Notations; Postfix Evaluation** Module 5 Enabling Assessment Formative Activity: Review for Summative Assessment 1/ Pointers to review	Schoolbook PowerPoint Canva	✓		2
	W9 Oct 14 – Oct 19	**MIDTERM SUMMATIVE ASSESSMENT** Coverage: Modules 1 to 5	Schoolbook		✓	2
			TOTAL	8	10	18

LEARNING PLAN FOR FINAL PERIOD:

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Apply the most appropriate data structures given a specified real world problem.	**TLO8.** Apply various applications of queue operations to real life scenarios. **TLO9.** Identify the different terminologies used in trees. **TLO10.** Explore different applications of binary trees. **TLO11.** Discover the varied ways of graph representations and applications. **TLO12.** Determine the steps in creating recursive routines.
CLO5. Create codes written in Java for various algorithms and data structures.	**TLO13.** Use and apply Java structure and elements to implement the algorithms and data structures covered in this course.
CLO6. Demonstrate 21st century skills inall learning activities.	**TLO7.** Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

LEARNING OUTCOMES FOR FINAL PERIOD

Module Title and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy	Technology Enabler	Asynchronous	Onsite	Allotted Hours
Module 6 TLO8, CLO4, CLO5, CLO6	**W10** Oct 21 – Oct 26	Gospel Reading, Reflection and Prayer **Module 6: Queue as an abstract data type** Class Participation/ Warm-Up Activity: Using Wooclap for Module 6	DLSU-D Schoolbook Newsfeed		✓	2
	W11 Oct 28 – Nov 2	**Independent Learning Week**	DLSU-D Portal	✓		2
Module 7 TLO9, CLO4, CLO5, CLO6	**W12** Nov 4 – Nov 9	Gospel Reading, Reflection and Prayer **Module 7: Tree as an abstract data type** Class Participation/ Warm-Up Activity: Using Wooclap for Module 7	Schoolbook PowerPoint		✓	2
Module 8 TLO10, CLO4, CLO5, CLO6	**W13** Nov 11 – Nov 16	Gospel Reading, Reflection and Prayer **Module 8: Binary Tree as an abstract data type** **Enabling Assessment: Quiz on Queue and Tree**	Classroom PowerPoint Canva	✓		2
Module 8 TLO10, CLO4, CLO5, CLO6	**W14** Nov 18 – Nov 23	Gospel Reading, Reflection and Prayer **Module 8: Trees and Binary Trees' Applications** Enabling Assessment: Quiz on Binary Trees	Schoolbook PowerPoint Quizziz		✓	2

Module 9 TLO11, CLO4, CLO5, CLO6	**W15** Nov 25 - Nov 30	Gospel Reading, Reflection and Prayer **Module 9: Graph as an abstract data type** Class Participation Activity: Using Quizlet for Module 9 Enabling Assessment: Quiz on Graph	Schoolbook PowerPoint Canva	✓		2
Module 9 TLO11, CLO4, CLO5, CLO6	**W16** Dec 2 – Dec 7	Gospel Reading, Reflection and Prayer **Module 9: Graphs' Applications** Class Participation on Graphs Application	Schoolbook Quizlet		✓	2
Module 10 TLO12, CLO4, CLO5, CLO6	**W17** Dec 9 – Dec 14	**COMPLETION / CONSULTATION WEEK** *Students will comply with lacking assessments and/or in-person/online consultation with the professor*	Schoolbook PowerPoint Canva	✓		2
	W18 Dec 16 – Dec 18	**FINAL SUMMATIVE ASSESSMENT** Coverage: Modules 6 to 10	Schoolbook PowerPoint Canva		✓	2
		TOTAL		8	10	18

GRADING SYSTEM

Midterm

Enabling Assessments (EA)	-	50%
Class Participation (CP)	-	20%
Summative Assessments (SA)	-	30%
Total		100%

Finals

Enabling Assessments (EA)	-	50%
Class Participation (CP)	-	20%
Summative Assessments (SA)	-	30%
Total		100%

Overall Final Grade = [(Midterm grade) + (Final term grade)] / 2

Note: Class participation (CP) refers to your active engagement in class from day 1 to the last day of each term (midterm or final). CP includes, but not limited to, the following: actual attendance, observance of netiquettes (synchronous) and etiquettes, promptness in submitting tasks/assessments, relevant questions asked, participation in collaborative tasks, etc.

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 15% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

ONLINE REFERENCES

Call number or e-provider	Reference Material
e-Book	John Bullinaria 2019. Data Structures and Algorithms - School of Computer Science. Universityof Birmingham, UK. [accessed 2020 August 10] https://www.cs.bham.ac.uk.
e-Book	Data Structures and Program Design Using Java : A Self-Teaching Introduction Published in: 2020 Database:eBook Academic Collection (EBSCOhost) By: Malhotra, D.;Malhotra, N.
e-Book	Learn Data Structures and Algorithms with Golang : Level up Your Go Programming Skills to Develop Faster and More Efficient Code Published in: 2019 Database:eBook Academic Collection (EBSCOhost) By: Bhagvan Kommadi
e-Book	Goodrich, M. T. Data Structures and Algorithms in Java. Department of Computer Science University of California, Irvine. John Wiley & Sons, Inc. Retrieved from https://enos.itcollege.ee/~jpoial/algorithms/GT
YouTube Channel	Computer Education or all. 2018. Data Structures and Algorithms Complete Tutorial Computer Education for All [accessed 2022 August 15] https://www.youtube.com/watch?v=YWnBbNj_G-U
YouTube Channel	Tech Course PH. 2020. Data Structures and Algorithms Lecture Series. [accessed 2022 August 15] https://www.youtube.com/watch?v=CNhj4GZ_7iE

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